CalSimPy:

Making the CalSim3 Data-Handling Process a Little Easier, One Line of Python Code at a Time

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Outline

- Motivation & Core Concepts
- Example CalSim functionality
- Extending to the C2VSim groundwater model
- … and CalSimHydro, too
- Application: CalSimPy and resequencing hydrology
- Sharing and next steps

Motivation

- A single CalSim3 simulation entails a huge amount of disparate data
  - Meteorology, land use, streamflows
  - CalSimHydro, CalSim3, C2VSim
  - DSS, ascii tables, CSVs, GIS
- Excel spreadsheets are great for some of these things, but can be limiting when working with multiple types of data or many CalSim studies
- What if we had a flexible code library that made it easier to work with all of the different CalSim data?
  - My collection of *ad hoc* Python scripts → CalSimPy

CalSimPy – Core Concepts

- Each “model” or component is an object (a Python class):
  - A container to hold all the configuration and input/output data
  - Methods to interact with that data
- Access the model information the same way the numerical engines do (via *.launch, “main”, or “.in” files)
- Goal: make interacting with data across many models & many studies easier (to setup, to execute, to repeat, to adapt….)

CalSimPy: The CalSim object

- A container for CalSim study information
- Configuration
  - Start date – End Date
  - SV file path
  - DV file path
  - Init file path
  - DSS A part
  - DSS F part
  - *And all other info in the launch/config files*

```
<[1]>: cs3_study = cs3.calsim(launchFP = launch_fp1, calsim_version=3)

<[2]>: cs3_study

########################################
CalSim object:
        Version: CalSim3
        Study Directory:
D:/02_Projects/CalSim/proj/2023_CalClimateInit/calsim/CalSim3_Scenarios\s0002_DCR2023_9.3.1_danube_adj/Model_Files
        Main file: mainCS3_ReOrg_UWplusVF.wresl
        Configuration:
            Start Year-Month-Day: 1921-10-31
            End Year-Month-Day: 2021-9-30
            DV Filename:
DCR2023_DV_9.3.1_v2a_Danube_Adj_v1.8.dss
            SV Filename: DCR2023_SV_Danube_Adj_v1.8.dss
            DSS DV A Part: CALSIM
            DSS F Part: L2020A
########################################
```

CalSimPy: The CalSim object

(1) Initialize a CalSim object using a launch file path

(2) Check the CalSim object

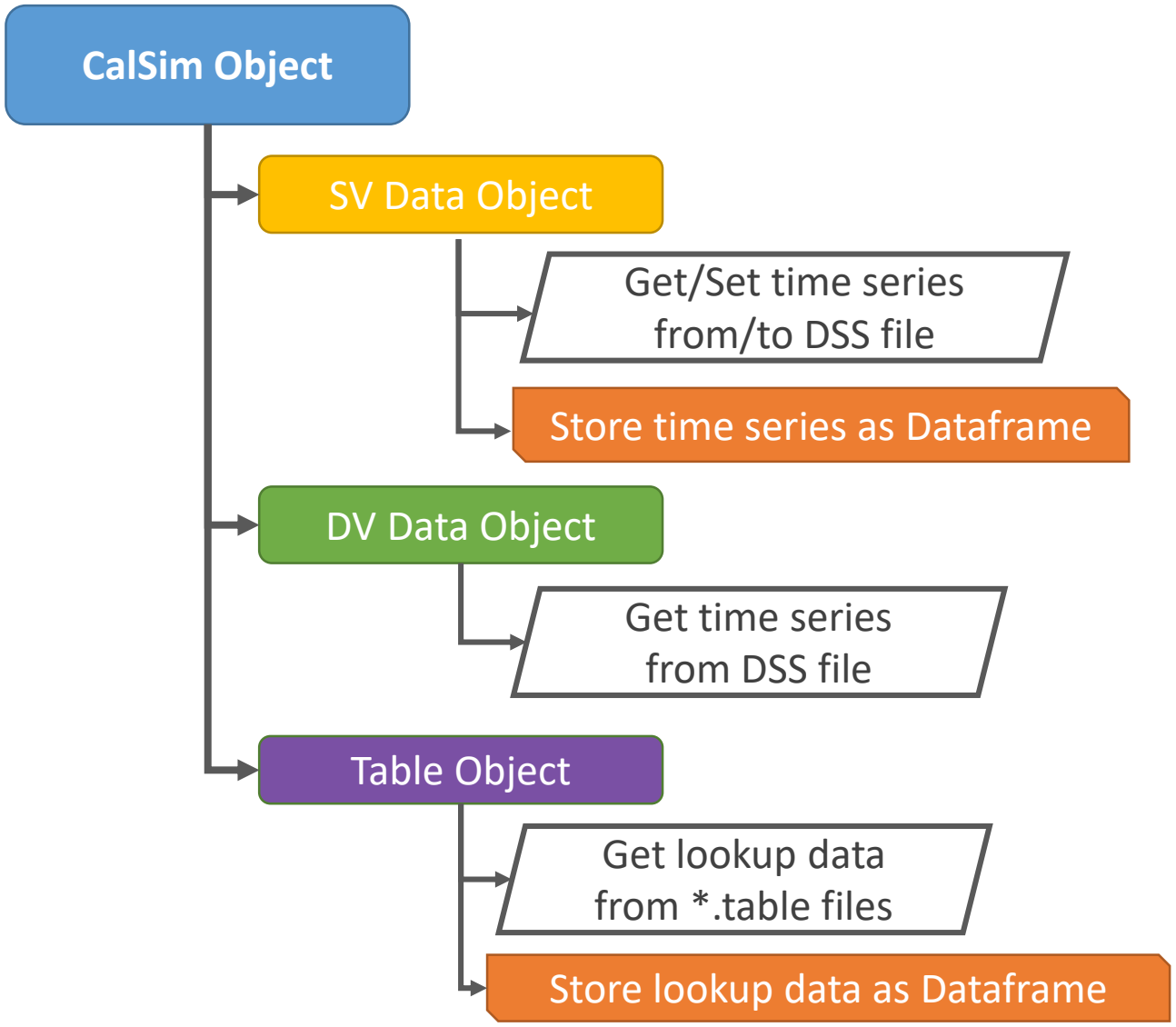
Response – subset of the study configuration info

```
<[1]>: cs3_study = cs3.calsim(launchFP = launch_fp1,
calsim_version=3)
```

```
<[2]>: cs3_study
```

```
########################################
CalSim object:
        Version: CalSim3
        Study Directory:
D:/02_Projects/CalSim/proj/2023_CalClimateInit/calsim/CalSim3_Sc
enarios\s0002_DCR2023_9.3.1_danube_adj/Model_Files
        Main file: mainCS3_ReOrg_UWplusVF.wresl
        Configuration:
            Start Year-Month-Day: 1921-10-31
            End Year-Month-Day: 2021-9-30
            DV Filename:
DCR2023_DV_9.3.1_v2a_Danube_Adj_v1.8.dss
            SV Filename: DCR2023_SV_Danube_Adj_v1.8.dss
            DSS DV A Part: CALSIM
            DSS F Part: L2020A
########################################
```

CalSimPy: CalSim object – Data



- Separate “containers” for SV and DV data
- Integrated DSS interface using *yapydss* library (“yet another Python DSS” wrapper)
- Table data read/write access as well (specify by table name)

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Example: Reading SV data from DSS

```
# get some SV data
bl_study.SVdata.getSVts(filter=['/I_SHSTA/'])  # Let's get the Shasta inflow time series

# We can also filter DSS pathnames using regular expressions
# the `append_dataframe = True` option will add the closure term
# data to the new dataframe rather than create a new dataframe (the default)
bl_study.SVdata.getSVts(filter=["CT_\S*_SV"], filt_method='regex', append_dataframe=True)

# all the retrieved timeseries data lives i
# be accessed using the `SVtsDF` attribute
svDF = bl_study.SVdata.SVtsDF

# we can confirm the data we have in the ne
svDF.head()
```

getSVts function uses filters (a list of partial matches or regular expressions) to select DSS paths

A	CALSIM						
B	I_SHSTA	CT_BENDBRIDGE_SV	CT_BUTTECITY_SV	CT_COLUSA_SV	CT_DAVIS_SV	CT_FAIROAKS_SV	CT_FREEPORT_SV
C	INFLOW	CLOSURE-TERM	CLOSURE-TERM	CLOSURE-TERM	CLOSURE-TERM	CLOSURE-TERM	CLOSURE-TERM
E	1MON	1MON	1MON	1MON	1MON	1MON	1MON
F	L2020A	L2020A	L2020A	L2020A	L2020A	L2020A	L2020A
Type	PER-AVER	PER-AVER	PER-AVER	PER-AVER	PER-AVER	PER-AVER	PER-AVER
Units	TAF	TAF	TAF	TAF	TAF	TAF	TAF
1921-31	231.689117	-3.560000	-37.980000	0.0	0.0	-1.63	0.000000
1-30	220.355118	-29.000000	-18.950001	0.0	0.0	0.00	21.170000
1-31	334.589569	-100.959999	-48.389999	0.0	0.0	0.00	59.660000
2-31	266.592133	-31.360001	17.520000	0.0	0.0	0.00	16.059999
1922-02-28	587.583557	9.910000	59.750000	0.0	0.0	0.00	2.690000

5 rows × 27 columns

Function stores retrieved data in an internal Pandas dataframe that retains the DSS header information

Example: Modifying and writing SV data to DSS

- Let's say you want to change some of the SV data (e.g. reduce February Shasta inflow by 10%) and create a new DSS file with a new variable
- First – you can modify the data using operations on your dataframe

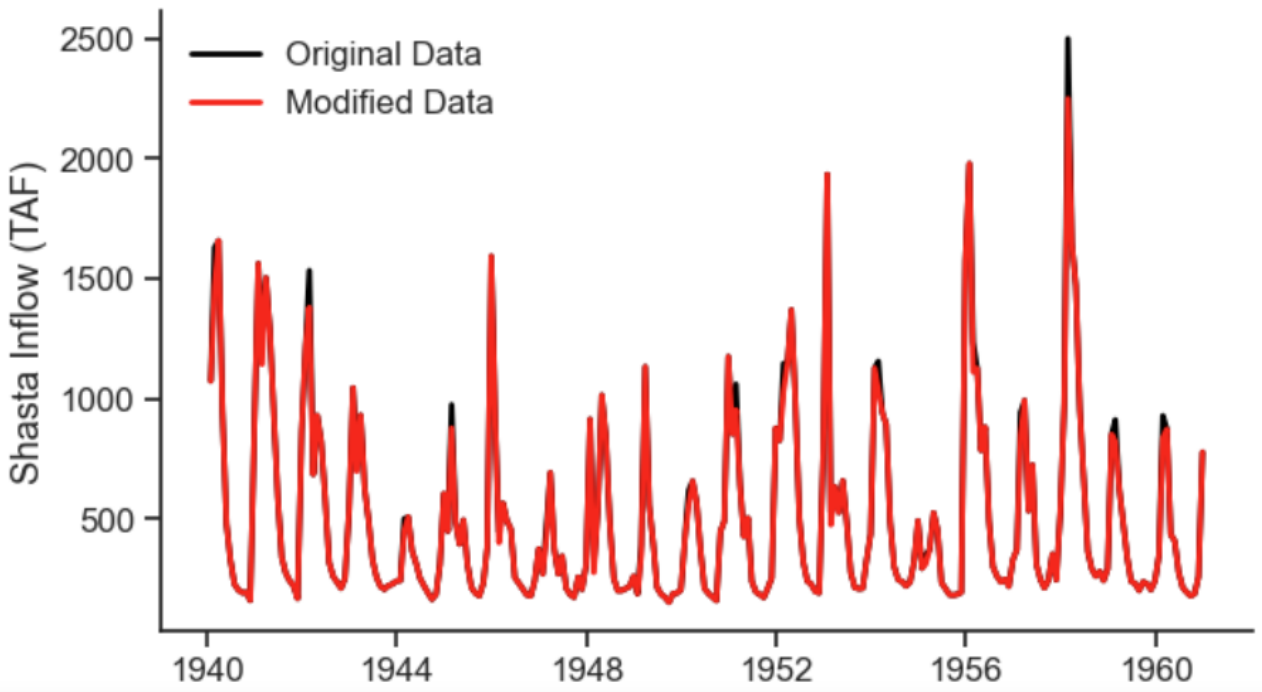
```
# let's reduce February inflow to Shasta by 10% as a test

# make a copy of your dataframe so we can make changes and not
# modify the original
svDF_mod = svDF.copy(deep=True)

# create a new variable
svDF_mod['CALSIM','I_SHSTA90pct','INFLOW','1MON','L2020A','PER-AVER','TAF'] =
svDF_mod.loc[:, idx[:,'I_SHSTA' ]].values

# filter out just the Februaries
feb_idx = svDF_mod.index.month==2

# do the calculation
svDF_mod.loc[feb_idx,idx[:,'I_SHSTA90pct']] = svDF_mod.loc[feb_idx,idx[:,
'I_SHSTA']].values*0.90
```

Example: Modifying and writing SV data to DSS

- Along the way you can make a nice plot to check that you did it correctly

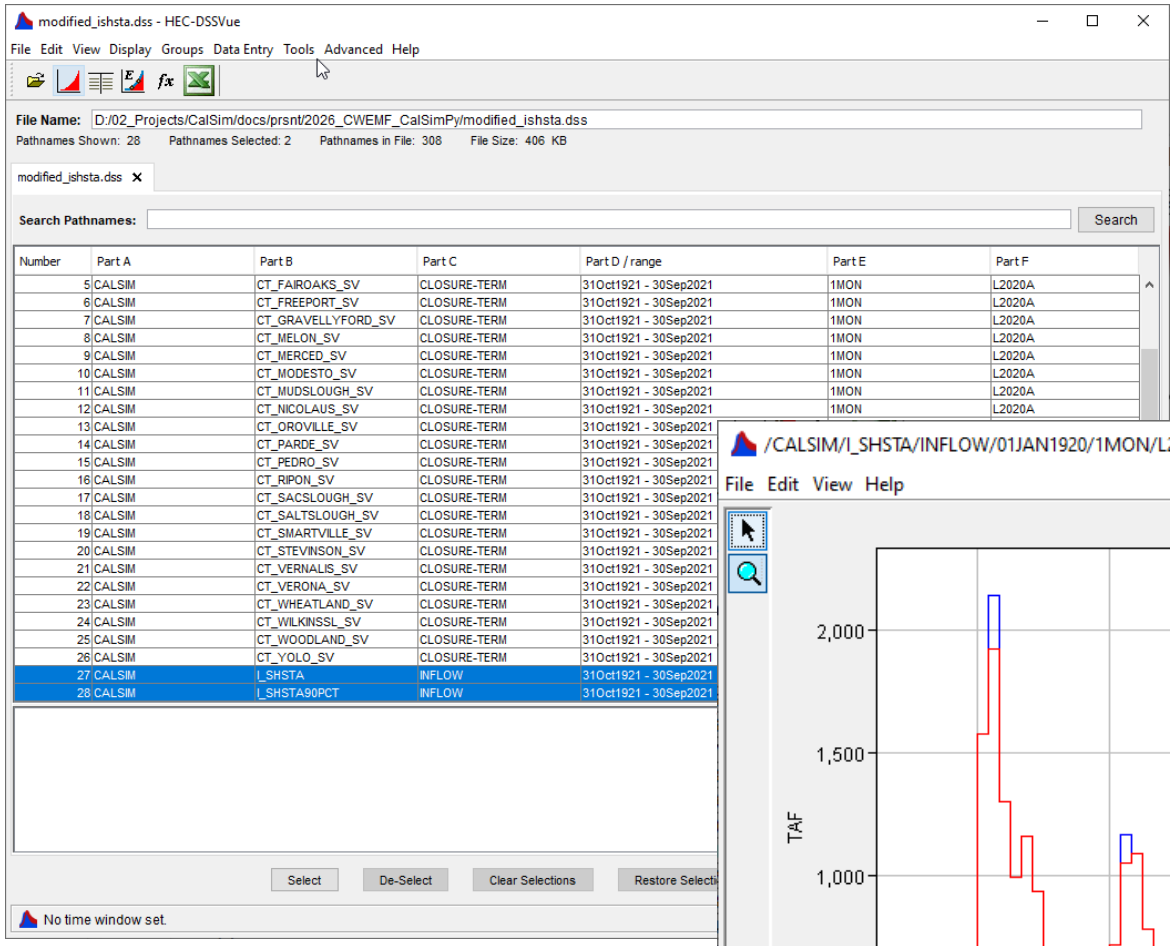
```
# we can plot this quick to check
with sns.plotting_context('notebook', font_scale=1.1):
    fig, ax= plt.subplots(1,1, figsize=(7,4))

    ax.plot(svDF_mod.loc['1940':'1960', idx[:, 'I_SHSTA']], c='k', lw=2,
    label='Original Data')
    ax.plot(svDF_mod.loc['1940':'1960', idx[:, 'I_SHSTA90pct']], c='red', lw=2,
    label='Modified Data')
    ax.set_ylabel('Shasta Inflow (TAF)')

    plt.legend(loc='best', frameon=False)
    sns.despine()
```

✓ 0.2s



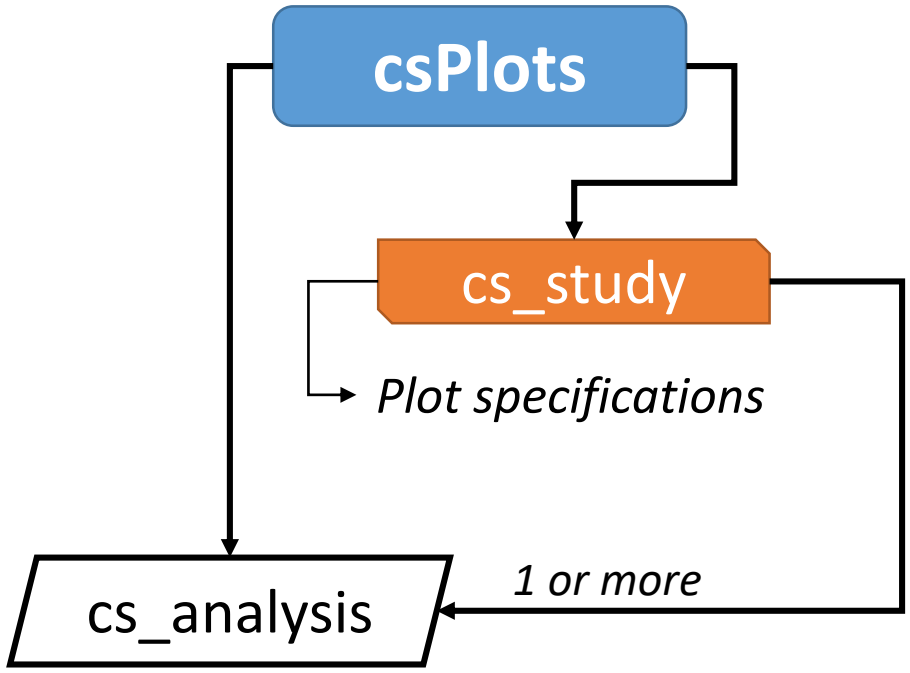
Example: Modifying and writing SV data to DSS

- Finally you can write the time series data to a new DSS file using the **setSVts** function



CalSimPy: Multi-study comparisons

- Often we want to compare data from multiple scenarios using different graphs and visualizations
- The set of functions in the **`csPlots`** module makes these sorts of comparisons easier
  - CalSim object → cs_study
  - cs_study → cs_analysis → data & plots!



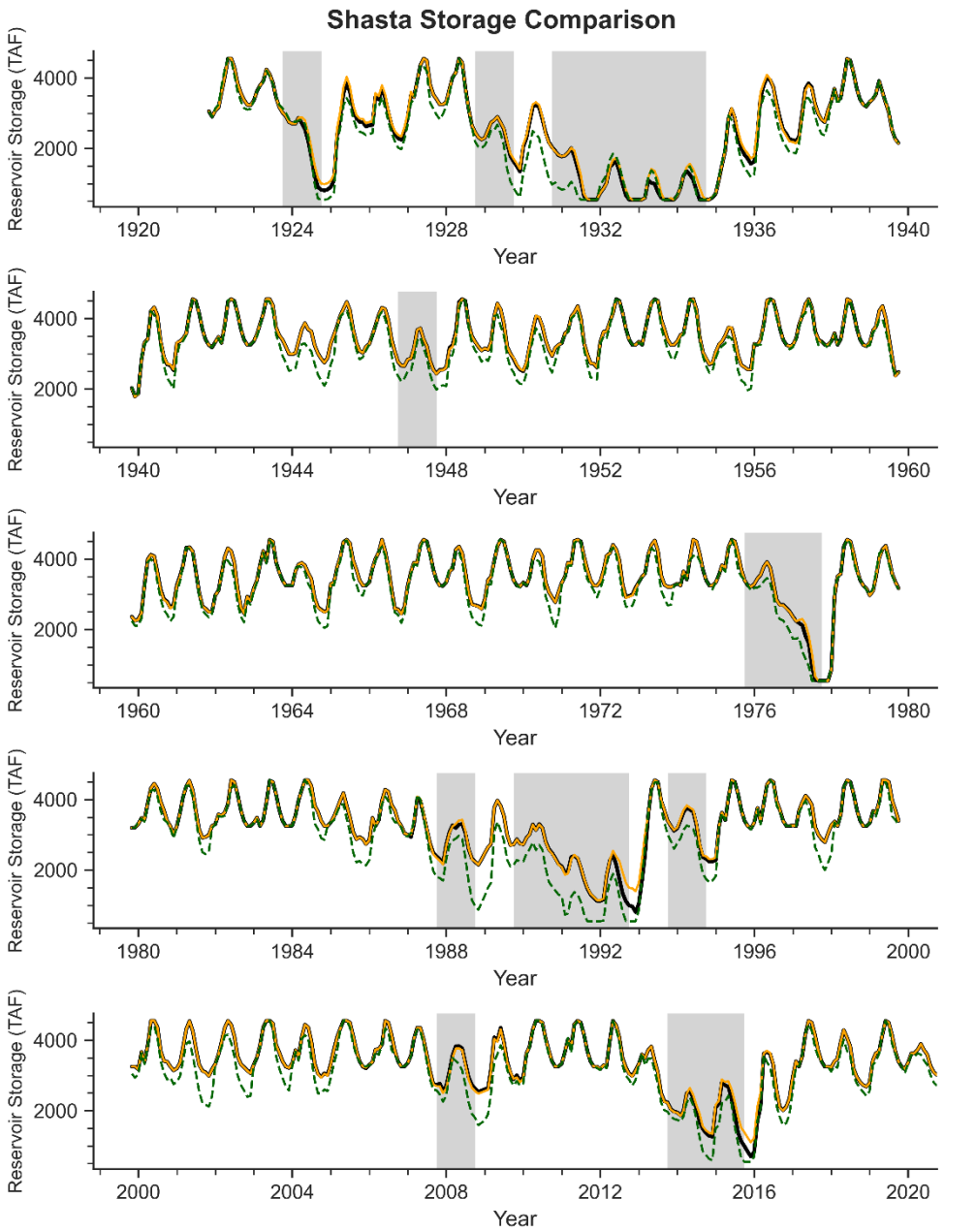
- *Retrieve data from DSS files*

- *Time series plots*
- *Exceedance plots*
- *Pattern plots*

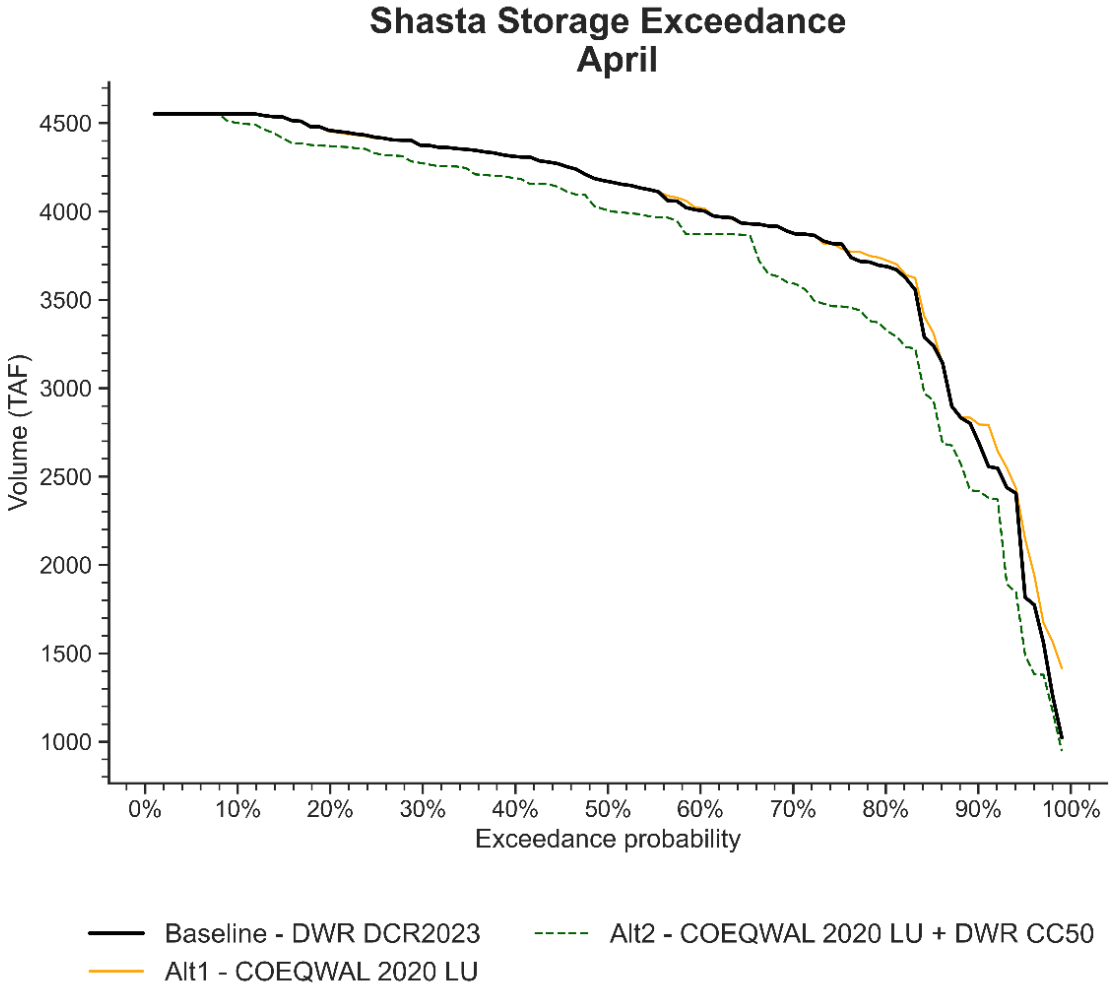
Example: Comparing 3 Scenarios – Time series

```python
# initialize other calsim studies

alt1 = cs3.calsim(configFP = config_fp2, calsim_version=3, reorg=True)
alt2 = cs3.calsim(launchFP = launch_fp3, calsim_version=3, reorg=True)

# create a cs_plots study for each - set a line color, name, and description
# line color and labels will be used across all plot types
bl_study_plot = csplt.cs_study(bl_study, "Baseline - DWR DCR2023",baseline=True,
                               color="k", desc="DWR's 2023 DCR baseline with
                               adjusted historical hydrology")

alt1_study_plot = csplt.cs_study(alt1, "Alt1 - COEQWAL 2020 LU",baseline=False,
                               color="orange", desc="COEQWAL version of 2023 DCR
                               baseline with adjusted historical hydrology and 2020
                               updated land use")
```

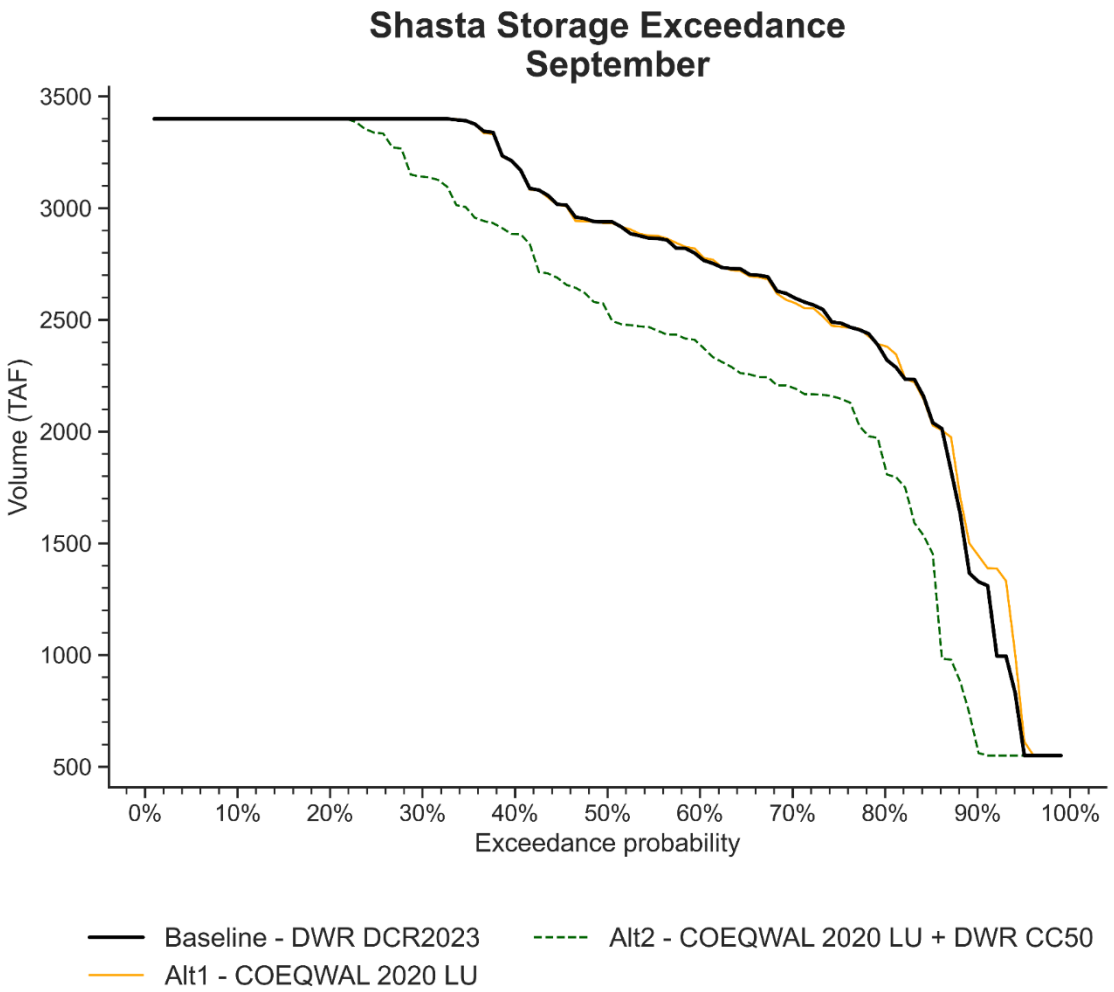
```python
#make an "analysis" with the studies
an1 = csplt.cs_analysis([bl_study_plot, alt1_study_plot, alt2_study_plot])

# get data for all scenarios in the study
an1.getDV(filter=['/S_SHSTA/','/DEL_CVP_PAG_S/'])

# set which variable you want to plot
an1.stageVar('S_SHSTA', 'DV')

# call the `plot_multi_monthlyTS` time series plotting function
fig_dict = an1.plot_multi_monthlyTS('Shasta Storage Comparison',
highlight_crit=True)
```



Example: Comparing 3 Scenarios – Exceedance - Select Months




```
fig_dict = an1.plot_annual_exceed('Shasta Storage', month_filter=[4],
                                  metric_axis=False,
                                  annualize_on='A-Sep')
```

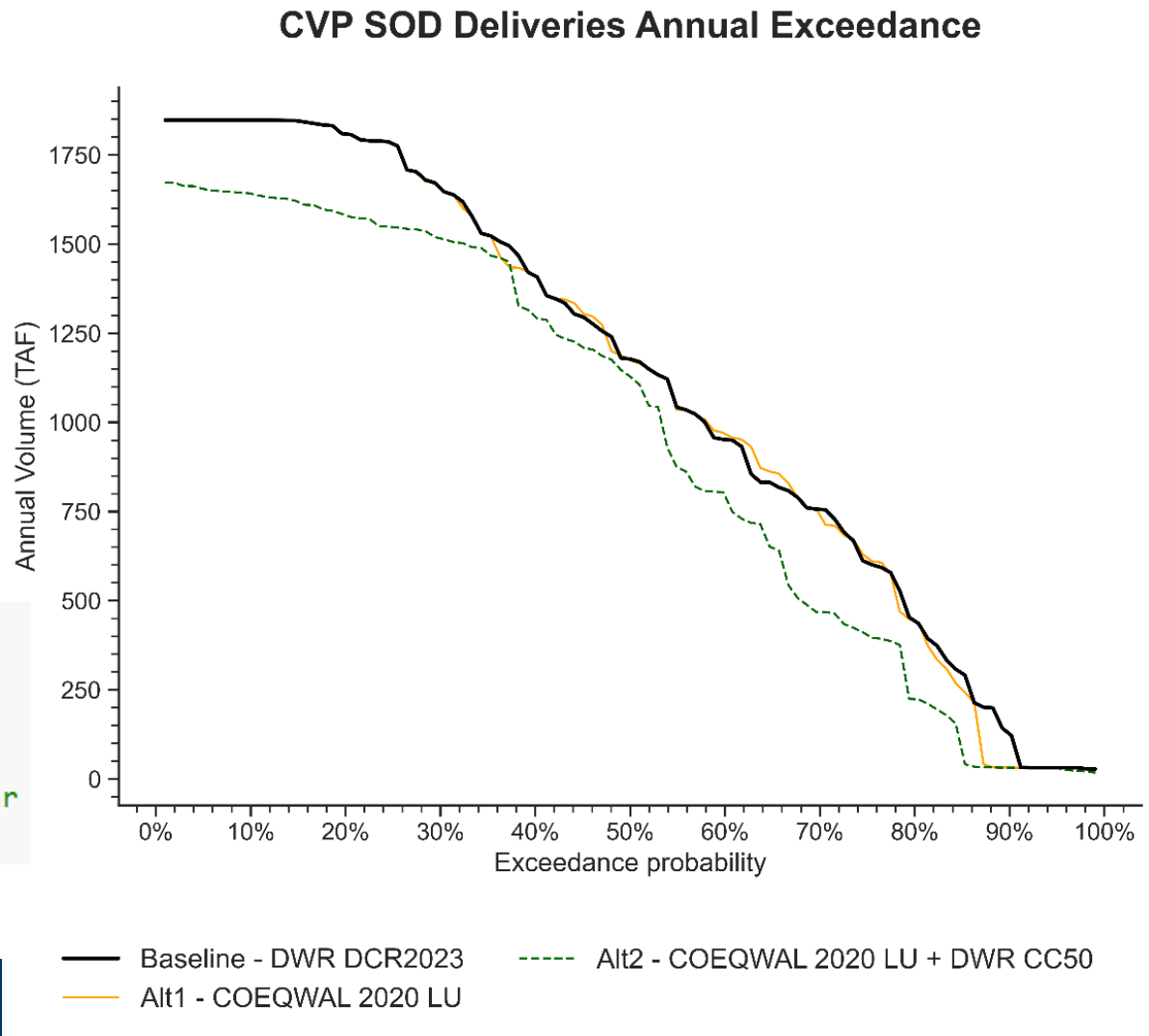



```
fig_dict = an1.plot_annual_exceed('Shasta Storage', month_filter=[9],
                                  metric_axis=False,
                                  annualize_on='A-Sep')
```

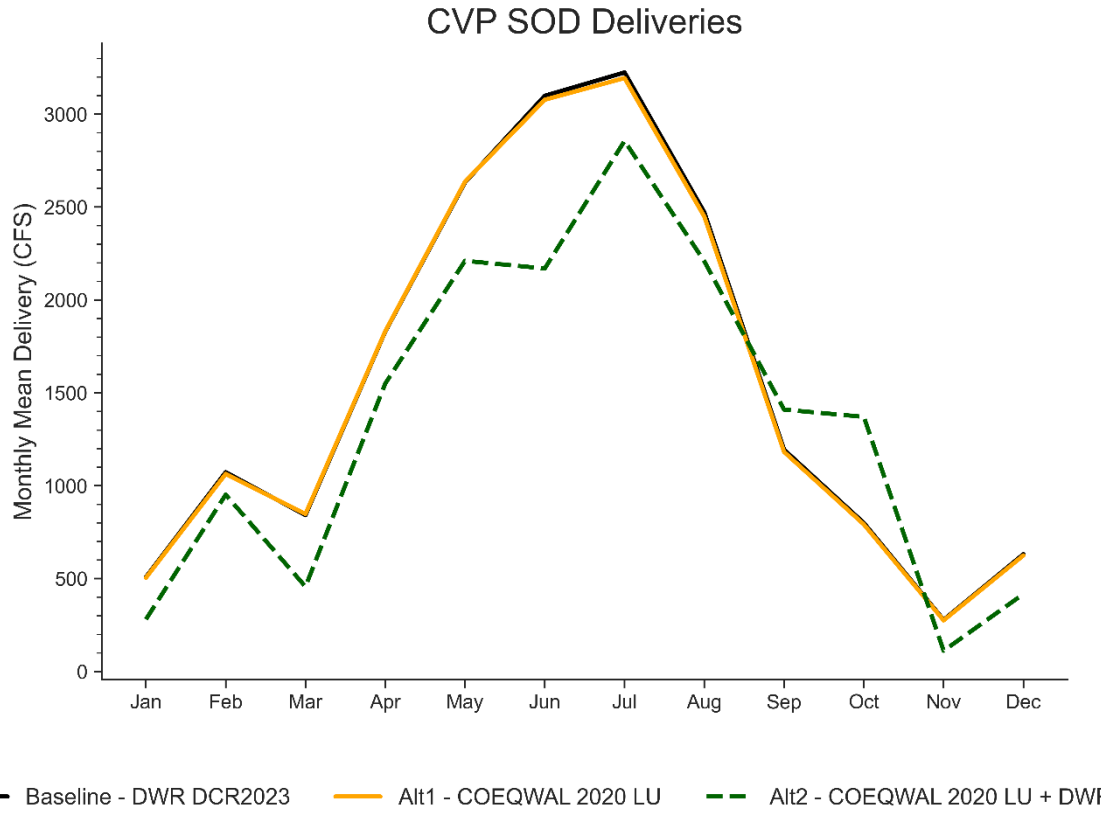
Example: Comparing 3 Scenarios – Exceedance - Annual Volumes

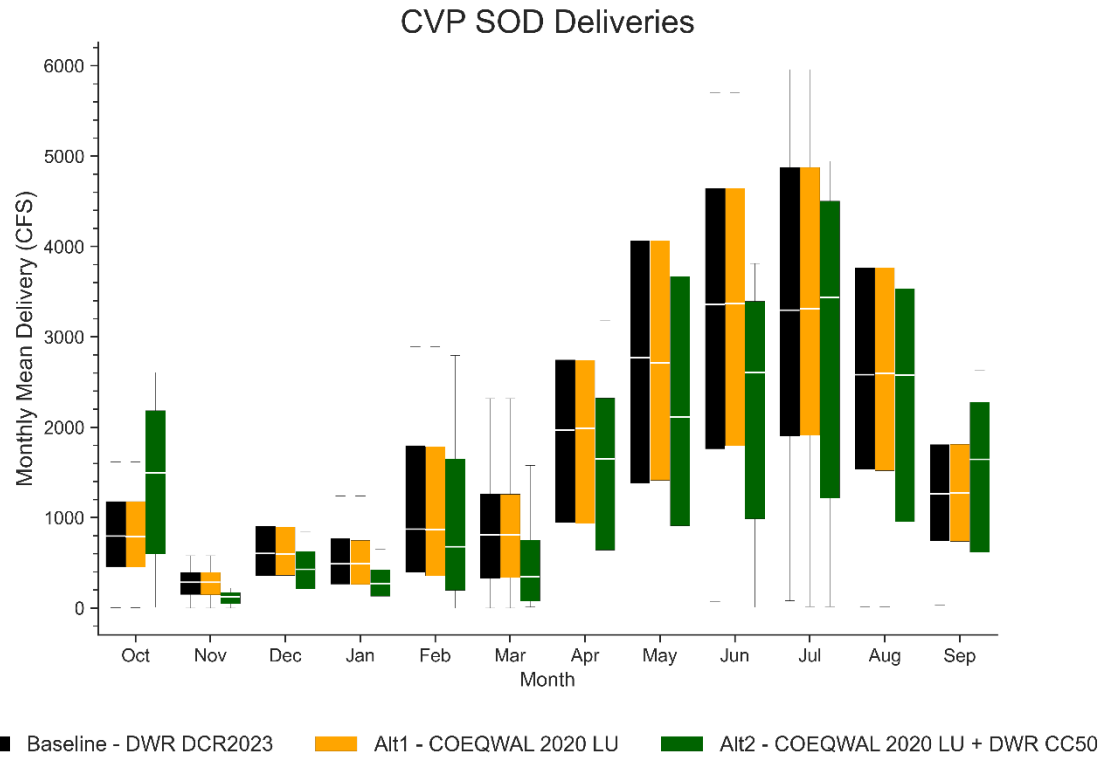
Automatically converts from CFS to TAF and aggregate to an annual period (e.g. Oct – Sep, Mar – Feb, Calendar Year)

```
an1.unstageVar('S_SHSTA')  #<-- unload the old variable
an1.stageVar('DEL_CVP_PAG_S', 'DV')  #<-- select a new variable

fig_dict = an1.plot_annual_exceed('CVP SOD Deliveries',
                                  annualize_on='A-Feb',  #<-- Contract year
                                  metric_axis=False)
```



Example: Comparing 3 Scenarios – Monthly Patterns

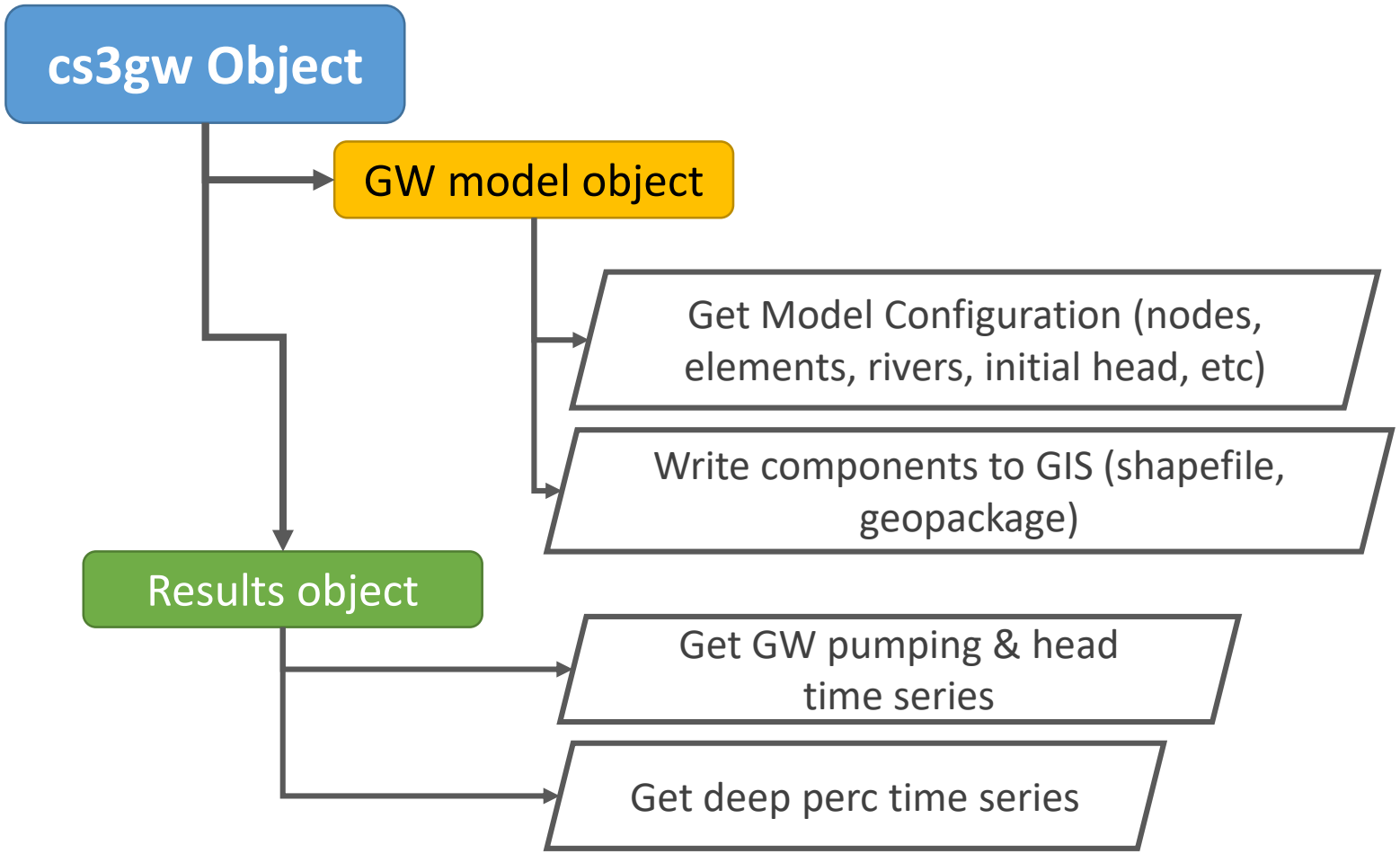





```
# Make a monthly pattern plot - use the average values, no filtering
fig_dict = an1.plot_pattern('CVP SOD Deliveries', 'Monthly Mean Delivery',
                            pattern_type='MarOct', annual_filter=[],
                            annual_filter_type="",
                            plotType='avg')
```

```
# Make a monthly pattern plot - show variability with box plot, no filtering
fig_dict = an1.plot_pattern('CVP SOD Deliveries', 'Monthly Mean Delivery',
                            pattern_type='MarOct', annual_filter=[],
                            annual_filter_type="",
                            plotType='box')
```

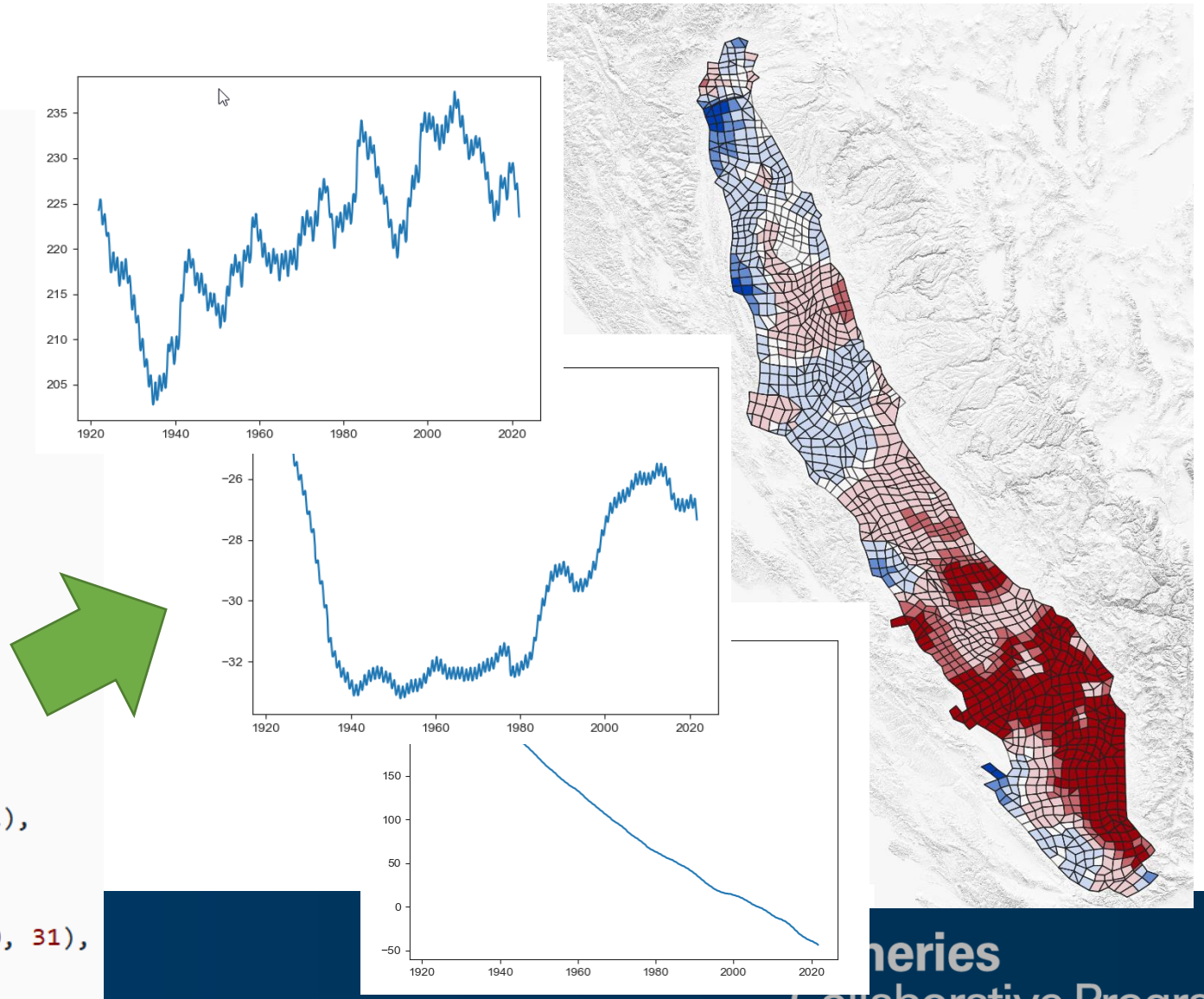
CalSimPy: Working with the C2VSim groundwater model



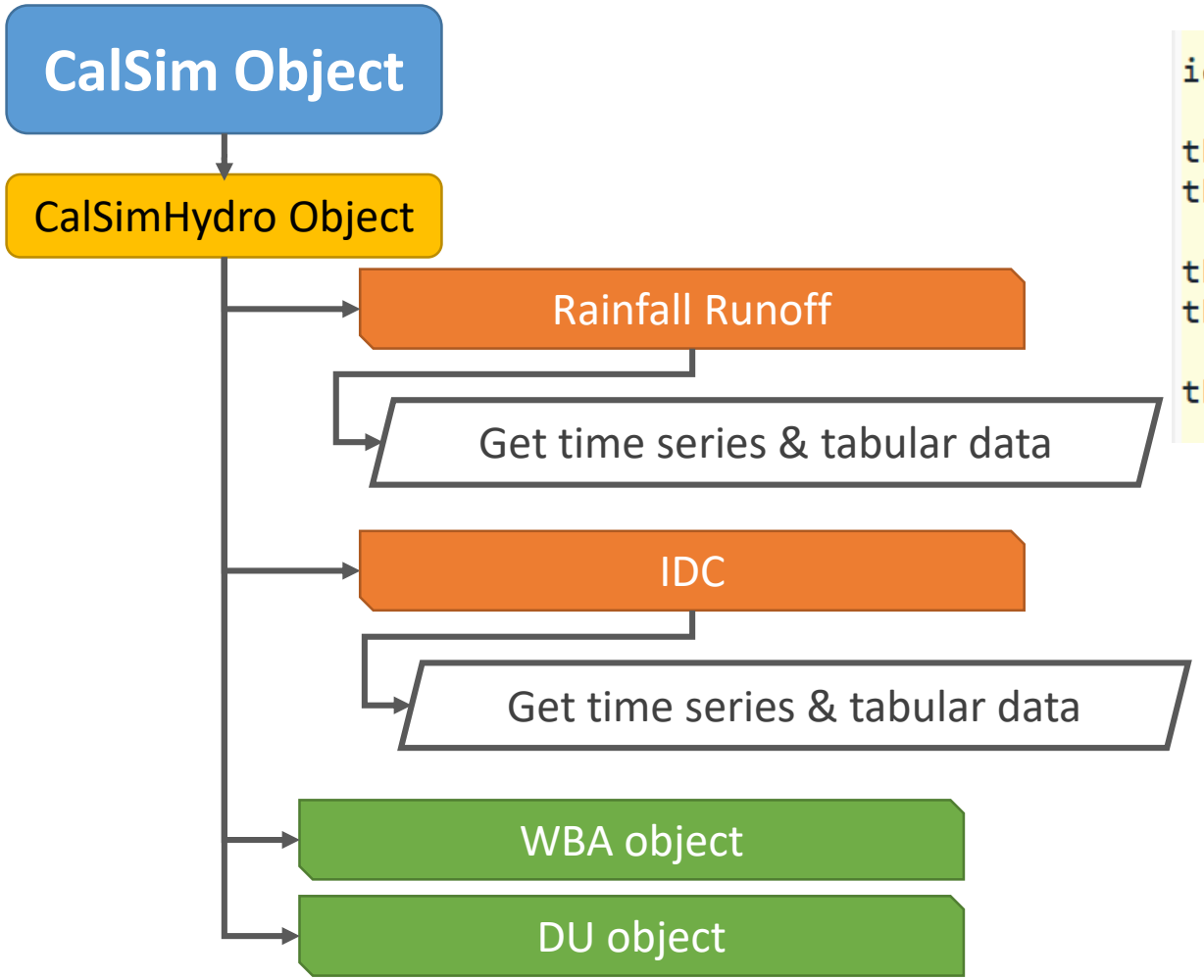
- Read in the C2VSim (CS3 DLL version) data via a cs3gw object
- Can get all the model components (boundary condition, nodes, elements, etc)
- Can read in the data from DSS, write spatial components to GIS

Example: Extract and Plot Groundwater Data as a Map

```
from calsimpy.cs3gw import cs3gw
from calsimpy.cs3gw import gwResults

import datetime as dt

gw1 = cs3gw(bl_study.projDir)

# read in the nodes and layers configuration
gw1.getNodes()
gw1.getStrat()

# read in the elements configuration
gw1.getElements()

# create a geodataframe of the elements
gw1.elem2GIS()

# get the geodataframe
elemGDF = gw1.ElementsGIS

# get the results
gw1_results = gwResults(gw1)

gw1_results.getGWHead(gw_startDate=dt.datetime(1921, 10, 31),
                      gw_endDate=dt.datetime(2021, 9, 30),
                      reorg=True)

gw1_results.getGWDeepPerc(gw_startDate=dt.datetime(1921, 10, 31),
                      gw_endDate=dt.datetime(2021, 9, 30),
                      reorg=True)
```



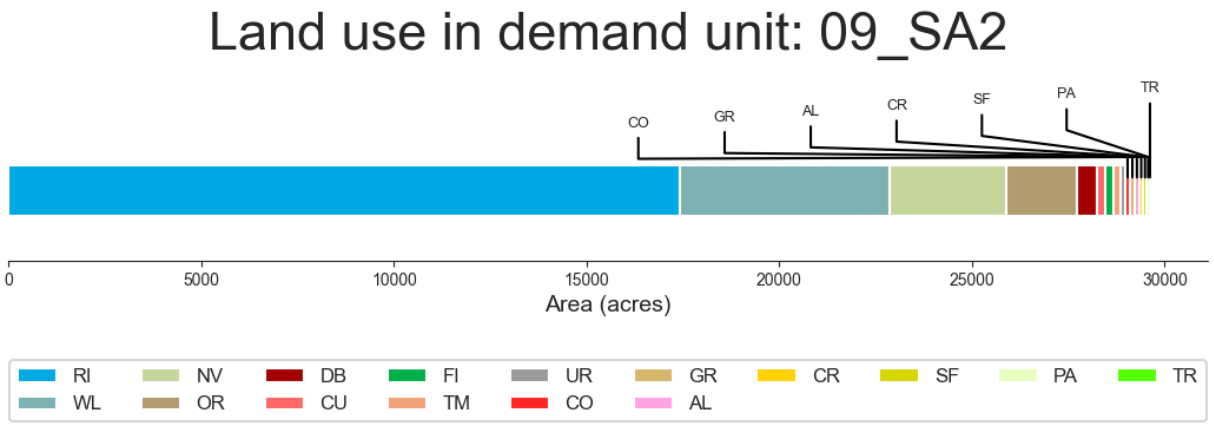
CalSimPy: CalSimHydro too!




```
idcfp = '/CalSimHydro/2a_IDC/Main.dat'

thiscsh = cs3.calsimhydro()  #<-- create a CSH object
thiscsh.IDC = cs3.idc(idcfp) #<-- initalize and IDC boject

thisIDC = thiscsh.IDC
thisIDC.init_DU_list() #<-- get DU info

thisIDC.get_land_use() #<-- retrieves land use for all DUs
```

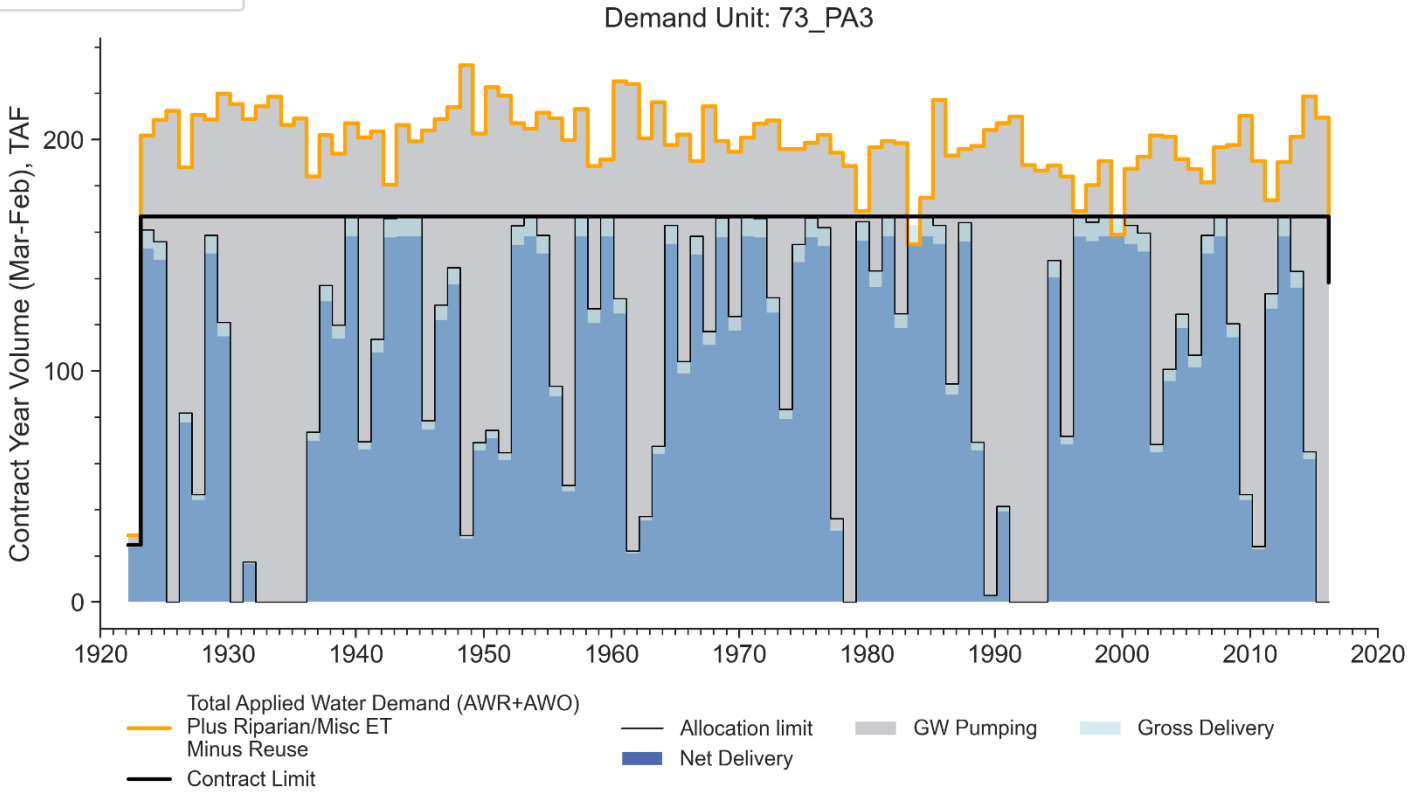
- Example use cases:
  - Easily adjust land use by DU (or whole domain)
  - View, analyze, or summarize meteorological inputs

Examples: CalSimHydro



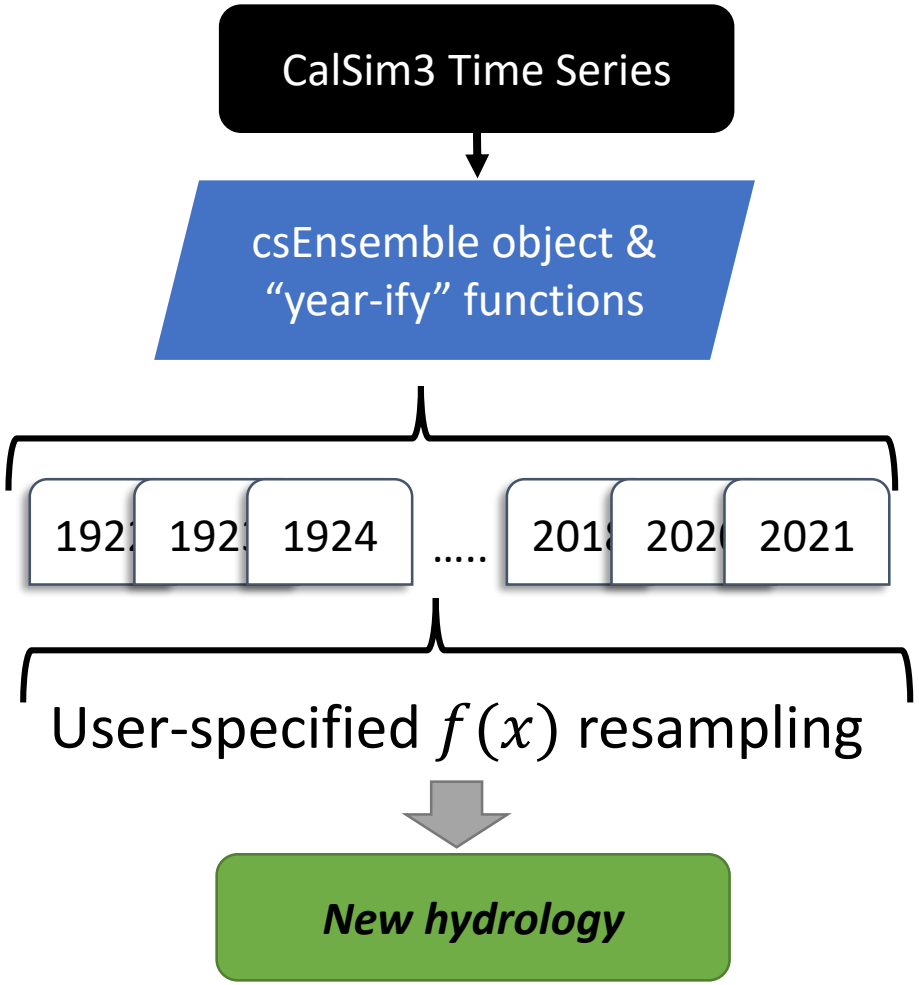
A land use summary for each Demand Unit

Combine CalSim3 and CalSimHydro data – delivery and water budget summary plots



Other applications: Custom hydrology

- Re-sequencing CalSim3 input hydrology
- Example – analysis completed for a project funded by California Department of Fish and Wildlife
- Concept – CalSim time series are split into yearly blocks
- Any custom sequence can be created



Example: Exploring novel drought sequences

- Synthetic annual sequence generated through a separate statistical method
- New selection of years becomes input to the code (e.g. [1987, 1943, 2018, 2014, 1977,….])
- **`shuffle_ts`** function handles the re-sequencing of CalSim yearly blocks

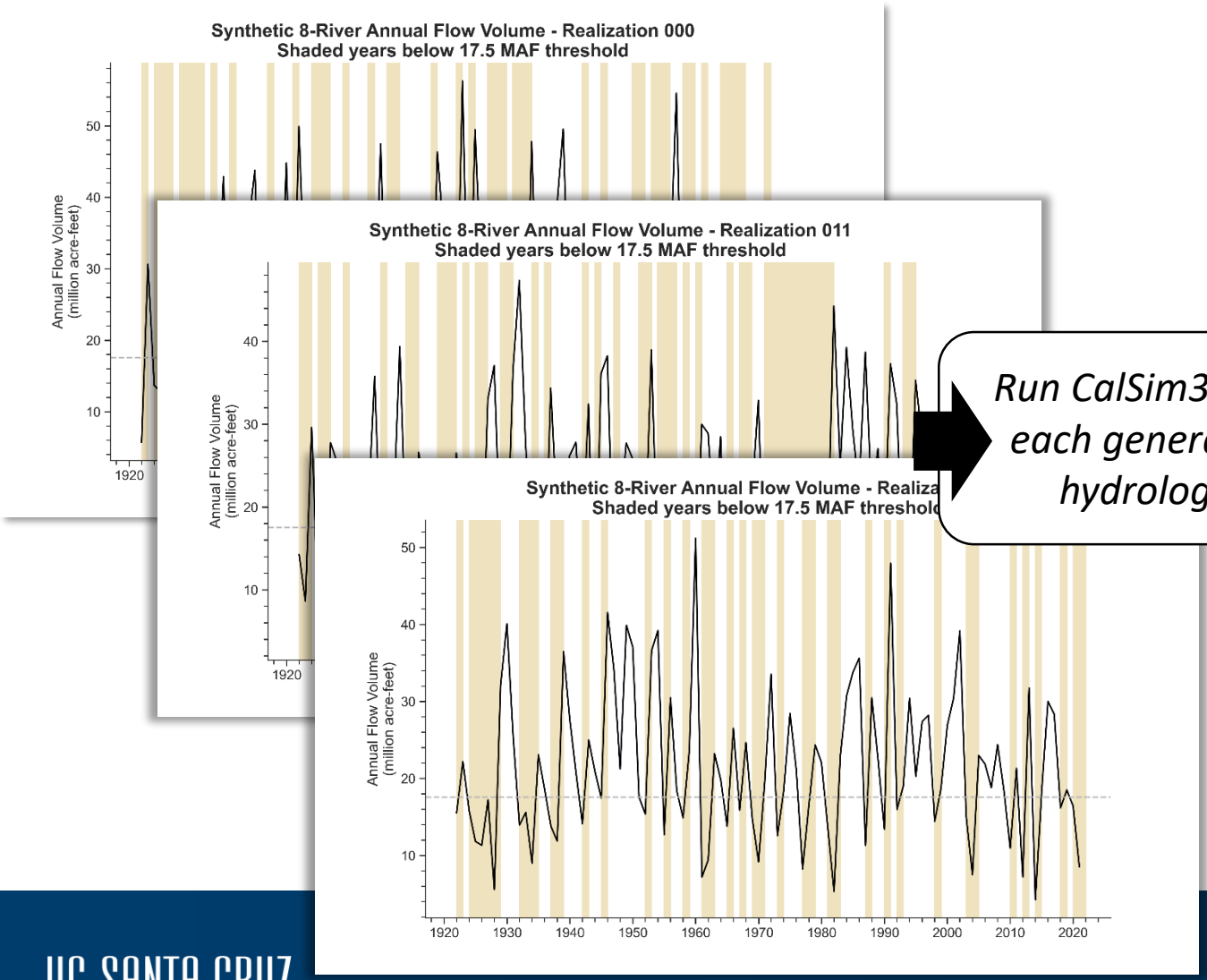
```
resampyr_fp = '/path/to/new/annual/hydrology/sequence.csv'

# load a CalSim3 base model
thiscs = cs3.calsim(base_launchFP)

# create an ensemble object from
# the CalSim model
ens1 = cs3.ensembleCS(thiscs)

# load in the new sequence of years
resamp_years_df = pnd.read_csv(resampyr_fp)
# convert it to a list
resamp_years_list = list(resamp_years_df.NewSequence)

# split the base data into yearly blocks
ens1.prepare_years()

# get a new dataset of re-ordered CalSim3 years
reordyr, reorddat, shufflyrs, seqYrs = \
    ens1.shuffle_ts(shuffle_list=resamp_years_list,
            exclude=[1919, 1920,1921,2022])
```

Example: Exploring novel drought sequences

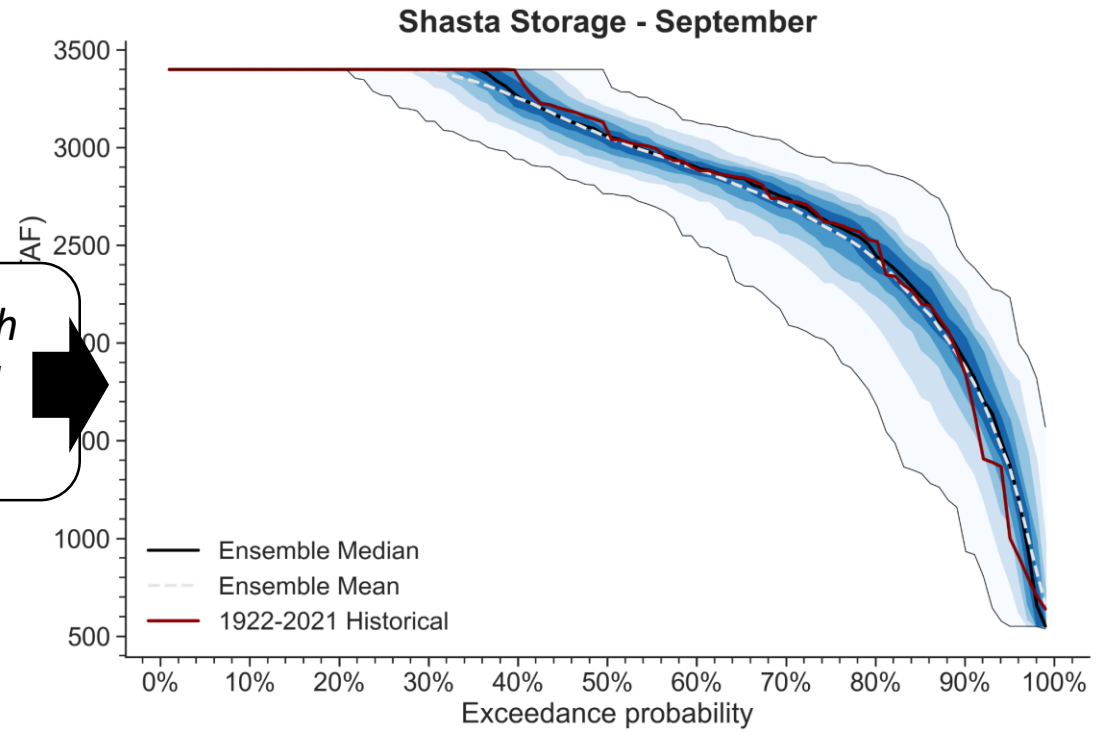
```
# combine the `reorddat` into a new dataframe
big_list = []
for yr, yd in zip(reordyr,reorddat): # loop through the keys
    #print(yr)
    if type(yd.Other) == type(None):
        dum =seqYrs.pop(seqYrs.index(yr))
    else:
        big_list.append(yd.Other)
seqDateIndex = pnd.date_range(dt.date(seqYrs[0]-1, 10, 31),
                              dt.date(seqYrs[-1], 9, 30),
                              freq='M')

newDF = pnd.concat(big_list, ignore_index=False, axis=0)
newDF.set_index(seqDateIndex, inplace=True)

# add the new data to the CalSim object,
# then write it to DSS
shuff01 = cs3.csSVdata(thiscs.LaunchFP)
shuff01.SVtsDF = newDF
shuff01.setSVts(newdssfp, debug=False)
```

- Combine the yearly blocks into a big dataframe again
- Set that as the SV object
- Write to a new SV DSS file
- Repeat as needed!

Example: Exploring novel drought sequences



Run CalSim3 with each generated hydrology



There is a lot left to do…

- Fill in missing functionality (e.g. with CalSimHydro)
- Improve consistency across components
- Documentation!

There is a lot left to do…but hopefully the code in its current form can be helpful

- CalSimPy code (warts and all!) is available on GitHub: **github.com/james-m-gilbert/calsimpy**
- A new round of updates coming soon
- Documentation & examples are also in the works
- Questions and constructive critiques welcome – james.gilbert@ucsc.edu

Thanks!